StataBridge

Universal Stata Data Converter

— Technical Documentation —

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1. Overview

StataBridge is a command-line data conversion tool designed for research teams that work across multiple versions of Stata. It solves the day-to-day problem of incompatible .dta file formats when collaborators are on different Stata releases — from Stata 7 all the way through Stata 19 — and extends that flexibility to four additional file formats commonly used in academic and applied research.

The tool was built to address three practical needs:

- **Format fragmentation:** Teams with mixed Stata versions (e.g. 13 and 18) cannot always exchange .dta files directly without data loss or read errors.
- **Data ingestion:** Stata researchers frequently need to import data from CSV exports, Excel files, or DBF files from GIS software and old database systems.
- **Data export:** Collaborators outside the Stata ecosystem may need data in CSV or Excel format.

StataBridge handles all of these scenarios through a single interactive CLI, with built-in safeguards for common failure modes such as long strings (strL), Unicode characters in old Stata formats, and illegal column names.

2. Key Features

2.1 Format Support

StataBridge reads and writes the following formats:

From \ To	Stata .dta	CSV	XLSX	DBF
Stata .dta	✓	✓	✓	✓
CSV	✓	–	✓	✓
XLSX / XLS	✓	✓	–	✓
DBF	✓	✓	✓	–
RData / rds	✓	✓	✓	✓

2.2 Stata Version Coverage

All thirteen Stata target versions are supported for .dta output. StataBridge maps each to the closest writable format supported by the underlying pandas library:

Version	Format	Writes As	Key Characteristics
Stata 7	108	114	Latin-1 encoding, str ≤ 244 chars
Stata 8	113	114	Latin-1 encoding, str ≤ 244 chars
Stata 9	114	114	Latin-1, str ≤ 244 — standard legacy target
Stata 10	114	114	Latin-1, str ≤ 244
Stata 11	114	114	Latin-1, str ≤ 244
Stata 12	114	114	Latin-1, str ≤ 244
Stata 13	117	117	strL introduced, Latin-1, str ≤ 2045 chars
Stata 14	118	118	Unicode (UTF-8), strL, str ≤ 2045
Stata 15	118	118	Unicode (UTF-8), strL, str ≤ 2045
Stata 16	119	119	Unicode, up to 32,767+ variables
Stata 17	119	119	Unicode, up to 32,767+ variables
Stata 18	119	119	Unicode, up to 32,767+ variables
Stata 19	119	119	Unicode — maximum pandas write version

Note: The pandas library supports writing formats 114, 117, 118, and 119 only. Stata 7–8 targets write format 114, which Stata 7 and later can read without issue. Stata 13 writes format 117 (the first format with strL support).

2.3 Automatic Safeguards

Several failure modes are handled automatically without requiring any user intervention:

strL Long-String Truncation

Stata introduced the strL variable type in version 13 (format 117), allowing strings up to 2 billion characters. Older formats (114 and below) cap strings at 244 characters. When converting to an older target format, StataBridge detects any column whose values exceed the target limit and truncates them, issuing a per-column warning.

```
Warning: Column 'abstract': truncated to 244 chars
         (target format 114 does not support strL)
```

Unicode to Latin-1 Transliteration

Stata formats 114 and 117 use Latin-1 (ISO 8859-1) encoding. Characters outside this range — such as Vietnamese, Arabic, Korean, or emoji — cannot be stored. When StataBridge detects such characters in data headed for an old format, it replaces them with the Latin-1 substitution character and issues a warning naming the column and the number of affected values.

```
Warning: Column 'city': 3 values had non-Latin-1 characters
         (replaced with '?') — format 114 does not support Unicode
```

Column Name Sanitization

Stata variable names must start with a letter or underscore, contain only alphanumeric characters and underscores, and be at most 32 characters long. StataBridge automatically sanitizes column names imported from CSV, Excel, or DBF files that do not meet these rules:

- `2020_revenue` → `v_2020_revenue` (digit-leading prefix added)
- Spaces and special characters replaced with _
- Names truncated to 32 characters with deduplication suffix if needed

3. Requirements

For now, StataBridge is a portable Windows executable file (.exe) in CLI mode. You just need to use Windows 7, 8, 10, or 11 to run in 64 bit mode application.

4. Usage

4.1 Language Selection

On first launch, StataBridge prompts for a UI language. The selection applies to all menu text, prompts, and status messages for that session. Supported languages are:

English	Indonesian (Bahasa)	Vietnamese (Tiếng Việt)	German (Deutsch)
French (Français)	Spanish (Español)	Italian (Italiano)	Portuguese (Português)
Japanese (日本語)	Chinese (中文)		

The language can also be changed at any time through the Settings screen.

4.2 Conversion Modes

Four conversion modes are available from the main menu:

Single File

Converts one file at a time. The user provides the input file path (drag-and-drop into the terminal is supported), selects the output format, the target Stata version (if outputting .dta), an optional output folder, and whether to overwrite existing files.

Multiple Files

Accepts a comma-separated list of file paths or glob patterns (e.g. *.dta, /data/project/*.csv). All matched files are converted in sequence using the same output settings.

Batch — Folder

Converts all supported files in a single folder. The user can optionally filter by file extension (e.g. only .dta files) and choose an output directory separate from the source folder.

Batch — Folder + Subfolders

Same as Batch, but recursively descends into subdirectories. The relative directory structure under the source folder is preserved in the output folder. For example:

```
Source:  /data/project/wave1/survey.dta
         /data/project/wave2/survey.dta
Output:  /export/wave1/survey.csv
         /export/wave2/survey.csv
```

4.3 Output Format and Stata Version

When Stata .dta is selected as the output format, a numbered list of all thirteen target versions is shown. Each entry displays the Stata version label, the underlying file format number, and a brief description of format capabilities. The current default is highlighted. The default version can be changed permanently in Settings.

When any non-Stata output format is selected (CSV, XLSX, DBF), the Stata version selection is skipped.

4.4 Settings

The Settings screen allows permanent session-level configuration of:

- UI language
- Default output format
- Default target Stata version

A built-in version reference table (also accessible from Settings) shows all supported Stata versions alongside their format numbers and key technical characteristics.

5. Technical Architecture

5.1 Module Structure

StataBridge is a single self-contained Python file (stata_converter.py) with no package structure required. Internally it is organized into four logical layers:

Layer	Responsibility
UI / CLI	ANSI colour rendering, menu navigation, progress bars, language strings
File Discovery	Path expansion, glob resolution, recursive directory walking, extension filtering
Conversion Engine	Reading, safeguard application (strL, Unicode, names), writing all formats
Language / Config	10-language string table, settings persistence within session

5.2 Read Pipeline

Every input file passes through the same pipeline regardless of source format:

- **1.** File extension is detected and the appropriate reader is called
- **2.** A pandas DataFrame is produced along with any available metadata (variable labels, value labels, dataset label) from .dta files
- **3.** The DataFrame is passed to the write pipeline

Reader details by format:

Format	Reader
`.dta`	pandas.io.stata.StataReader — preserves variable labels, value labels, and dataset label
`.csv`	pandas.read_csv with UTF-8 low_memory=False
`.xlsx`	pandas.read_excel with openpyxl engine
`.xls`	pandas.read_excel with xlrd engine
`.dbf`	dbfread.DBF if available; otherwise built-in pure-Python dBASE III/IV parser
`.rdata`	rpy2 with R load(); first data.frame in workspace is used
`.rds`	rpy2 with R readRDS()

5.3 Write Pipeline

For .dta output, the write pipeline applies three sequential transformation passes before calling the pandas writer:

Pass 1 — Column Name Sanitization

All column names are sanitized to comply with Stata naming rules. The sanitization algorithm:

- `re.sub(r'[^a-zA-Z0-9_]', '_', name)` — replaces any non-alphanumeric, non-underscore character with an underscore
- Prefixes names beginning with a digit with v_
- Truncates names to 32 characters
- Appends a numeric suffix (_1, _2, …) to resolve any duplicates after truncation

Pass 2 — strL Truncation

For target formats 114 and below (Stata ≤8 compatibility), all object-dtype columns are inspected. Any column whose maximum value length exceeds 244 characters is truncated using pandas string slicing, and a warning is recorded.

For target format 117 (Stata 13) the limit is 2045 characters. Formats 118 and 119 have no effective limit.

Pass 3 — Unicode Fallback

For target formats 114 and 117, all object-dtype columns are scanned for characters with Unicode code points above 255 (outside Latin-1). Any such column has its values re-encoded through Latin-1 with the 'replace' error handler, substituting unmappable characters with the Latin-1 replacement character. A per-column warning records how many values were affected.

Formats 118 and 119 are UTF-8 and skip this pass entirely.

5.4 Built-in DBF Reader

Rather than requiring dbfread as a hard dependency, StataBridge includes a pure-Python dBASE III/IV parser as a fallback. The parser reads the binary DBF header to extract field descriptors (name, type, length, decimal count), then iterates the fixed-width record area. Supported field types:

- `C` Character — decoded as Latin-1 string
- `N` / `F` Numeric / Float — parsed to int or float depending on presence of decimal point
- `D` Date — parsed to Python date object from YYYYMMDD string
- `L` Logical — mapped to Python bool (T/Y → True, F/N → False)

Deleted records (flagged with 0x2A in the deletion byte) are skipped automatically.

6. Limitations and Known Constraints

6.1 Stata Format Constraints

- **Format 108 (Stata 7) and format 113 (Stata 8)** cannot be written natively by pandas. StataBridge writes format 114 for these targets, which Stata 7+ can read without issues.
- **Format 115 (Stata 13 native)** is similarly not writable by pandas. Stata 13 is targeted via format 117, which Stata 13+ reads correctly.
- **Format 120 (Stata 19 native)** is not yet supported by pandas. Stata 19 is served via format 119, which Stata 16+ reads without issues.

6.2 Metadata Preservation

- Variable labels and value labels are preserved in Stata-to-Stata conversions.
- Variable labels are truncated to 80 characters for format 114 targets and 320 characters for format 117+.
- Value labels require format 117 or later; they are silently dropped when writing format 114.
- Non-Stata formats (CSV, XLSX, DBF) do not carry Stata metadata. Round-tripping through these formats loses labels.

6.3 DBF Output Constraints

- DBF column names are truncated to 10 characters (dBASE III specification limit).
- Only C (character), N (numeric), and D (date) types are written. Complex types are coerced to strings.
- Character column width is capped at 254 bytes.

6.4 R Data

- Requires R 4.0+ installed on the system.
- Requires rpy2 Python package.
- For .RData / .rda files, only the first data.frame found in the workspace is extracted.
- R output is not currently supported (StataBridge cannot write .rds or .RData files).

7. File Structure

```
stata_converter/
├── stata_bridge.exe    # Main application (single file, no package needed)
└── StataBridge_Documentation.md             # Quick-start guide
```

No installation step is required. Place stata_bridge.exe anywhere on the filesystem and it will run

8. Quick Reference

Common Workflows

Task	Steps
Downgrade dta to Stata 12	Single file → paste path → dta (Stata) → Stata 12 → Enter
Convert folder of CSVs to Stata 18	Batch folder → paste folder → filter: .csv → dta → Stata 18
Export dta to Excel for co-author	Single file → paste path → xlsx → output folder
Recursive batch, preserve structure	Batch + subfolders → source root → output root
Change default language	Settings → Change language → select
Check Stata format numbers	Settings → Stata Version Reference

Warning Messages Reference

Warning Text	Meaning and Action
`Column 'X': truncated to 244 chars`	Column had values over 244 chars; values clipped. Consider Stata 13+ target to preserve full strings.
`Column 'X': N values had non-Latin-1 chars`	Unicode characters replaced with '?'. Use Stata 14+ target for full Unicode support.
`Variable labels truncated to 80 chars`	Stata 12 and below allow only 80-char variable labels.

Appendix: Stata .dta Format History

The following table summarises the evolution of the Stata binary data format for reference:

Format	Introduced	Notable Changes
108	Stata 7	Original modern format; Latin-1 encoding; str max 244
113	Stata 8	Minor header changes; same string limits
114	Stata 9	Widely used legacy format; pandas read/write target for Stata 7–12
115	Stata 13	Introduced dataset notes; rarely produced by third-party tools
117	Stata 13	strL type (strings up to 2 billion chars); still Latin-1 for non-strL
118	Stata 14	Full Unicode (UTF-8) throughout; strL; 32,767 variable limit
119	Stata 15	Extended variable count beyond 32,767; otherwise same as 118
120	Stata 19	Minor extensions; not yet writable by pandas — format 119 used instead

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Built with Python 3.12 · pandas 3.0 · openpyxl 3.x